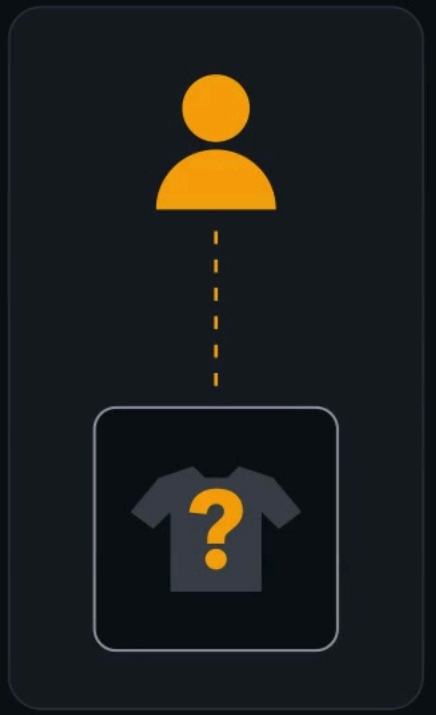
UDSIG

Generating Personalized Images for Sparse-Interaction Users with Uncertainty-Aware Retrieval and Dense Knowledge Guidance



Yuting Zhang · Ying Sun · Dazhong Shen · Ziwei Xie · Feng Liu
Changwang Zhang · Xiang Liu · Jun Wang · Hui Xiong

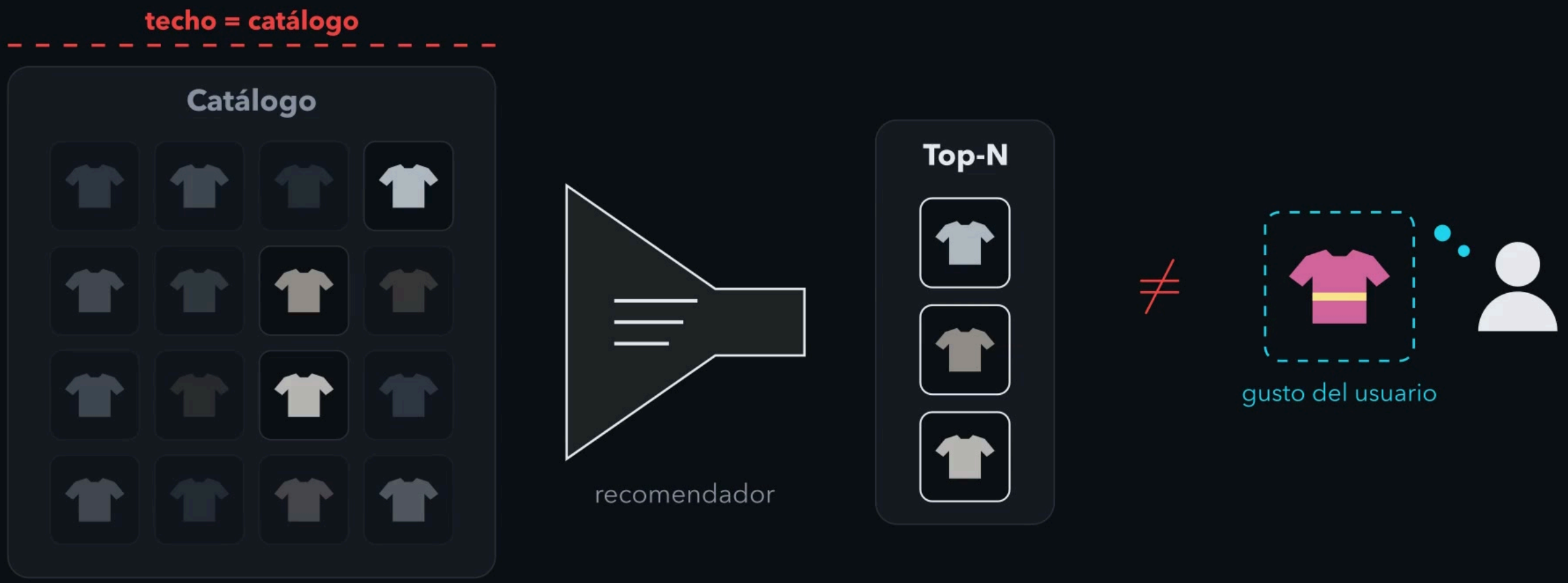
HKUST (Guangzhou) · NUDT · NUAA · OPPO

RecSys '26 · Minneapolis

doi.org/10.1145/3773078.3831767

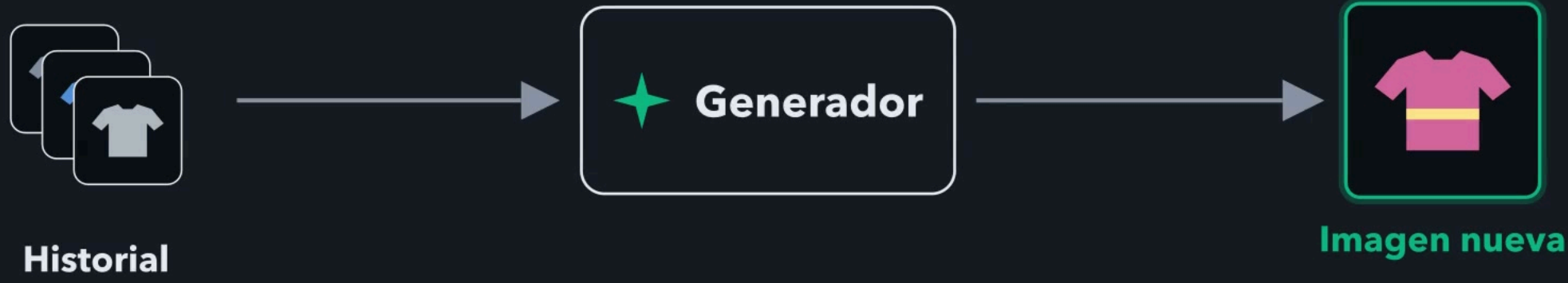
1 Contexto

Recomendar es filtrar



1 Contexto

Generar en vez de elegir



ya se usa en

Ads — *Chen X. et al. (2025)*

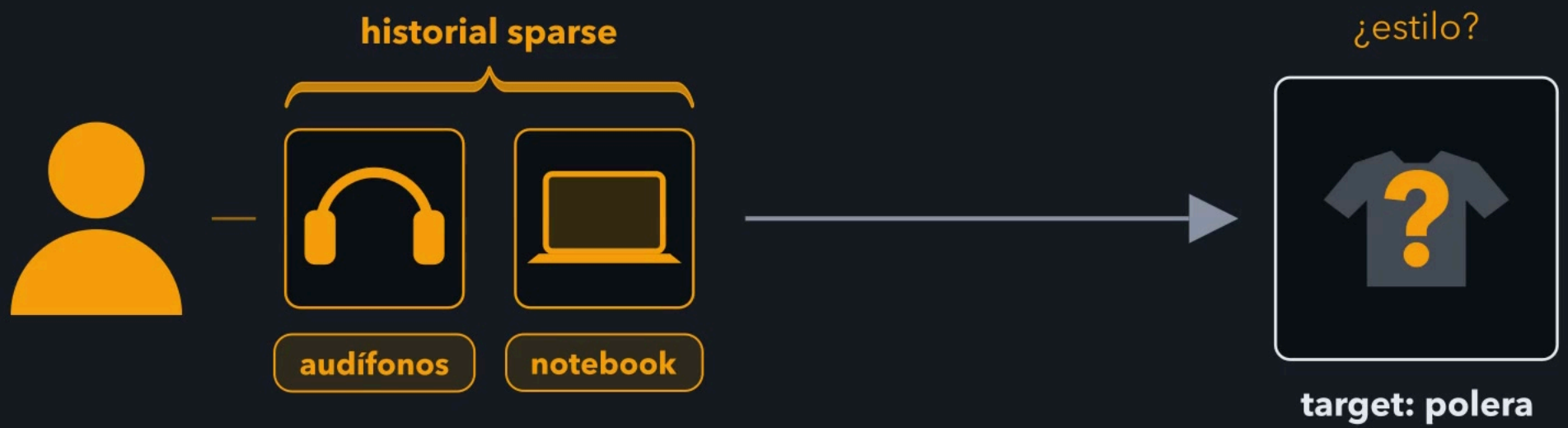
Posters — *Shen et al. (2024)*

Moda — *Xu et al. (2024)*

Zhang et al. (2026), Sec. 1

1 Contexto

Usuarios con poca interacción

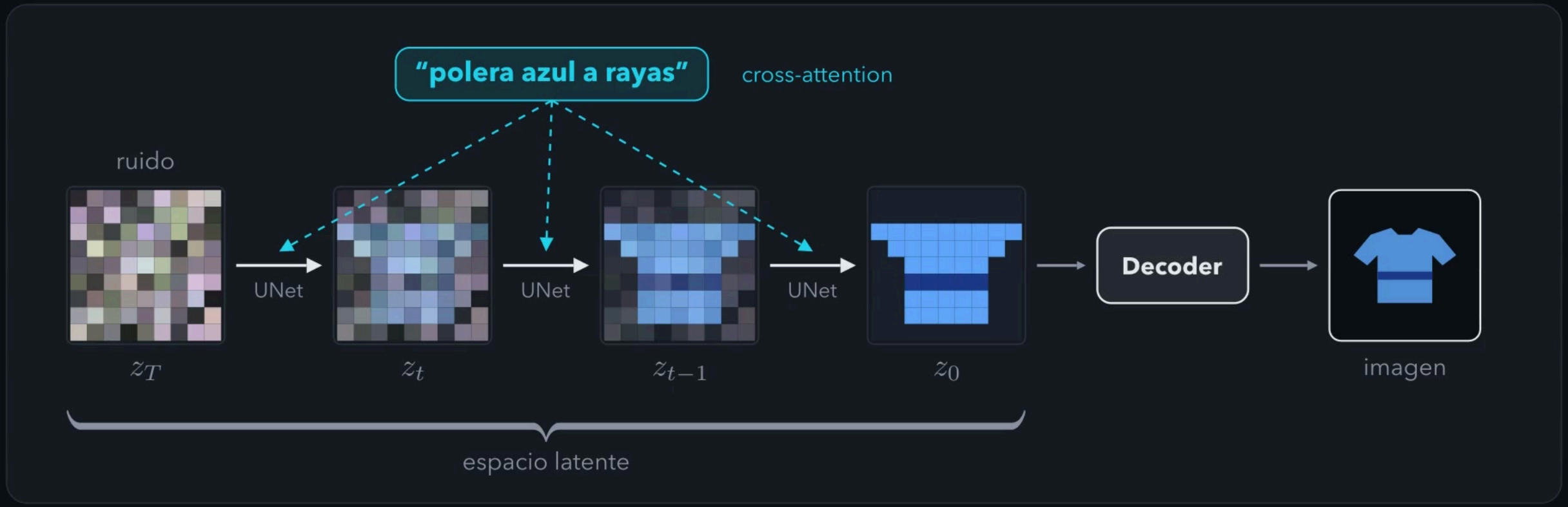


Chen Z. et al. (2025), data scarcity survey

2 Marco teórico

Cinco conceptos base

1 Difusión latente | 2 CLIP | 3 IP-Adapter | 4 Recomendadores | 5 Reward feedback



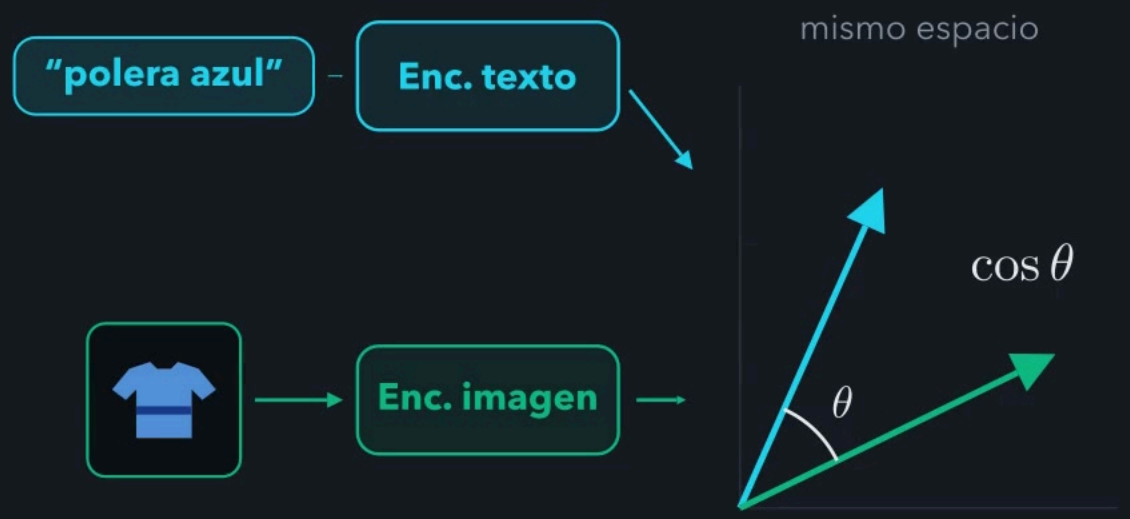
Rombach et al. (2022) · Podell et al. (SDXL)

2 Marco teórico

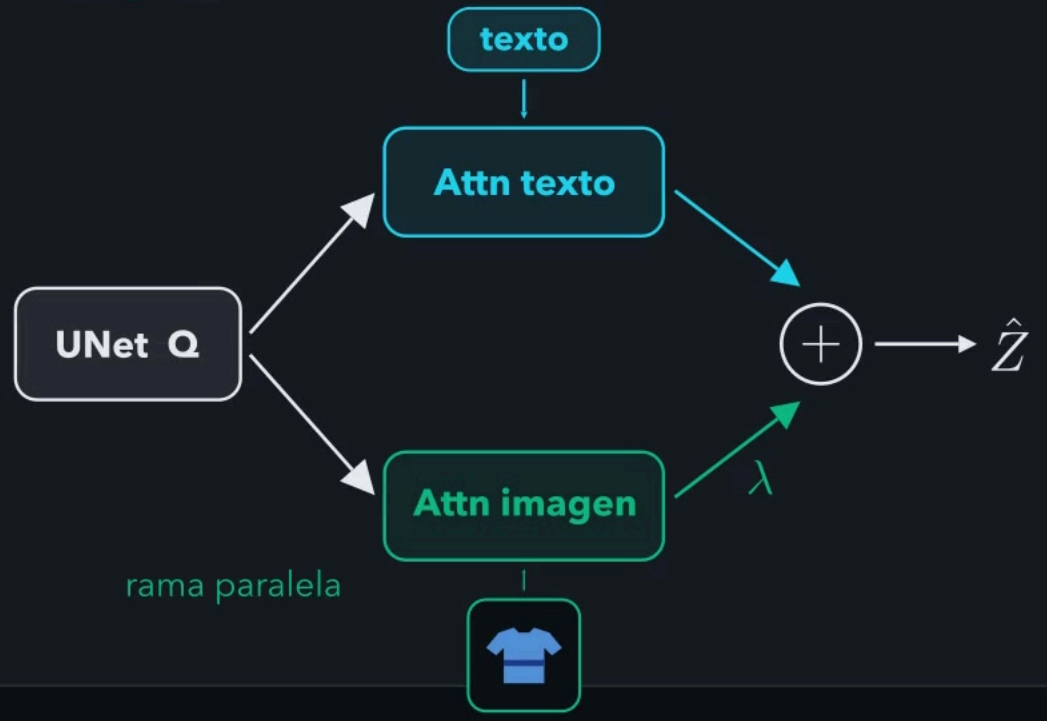
Cinco conceptos base

1 Difusión latente | 2 CLIP | 3 IP-Adapter | 4 Recomendadores | 5 Reward feedback

CLIP



IP-Adapter



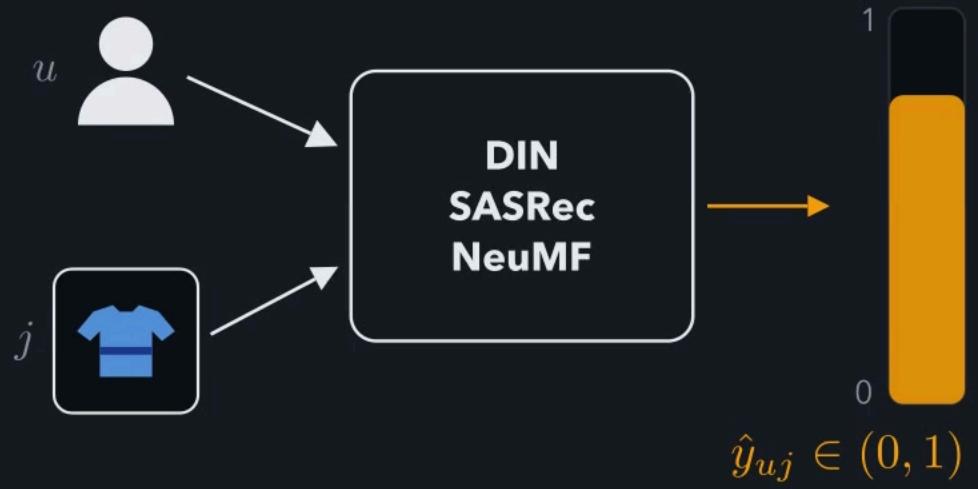
Radford et al. (2021) · Ye et al. (2023)

2 Marco teórico

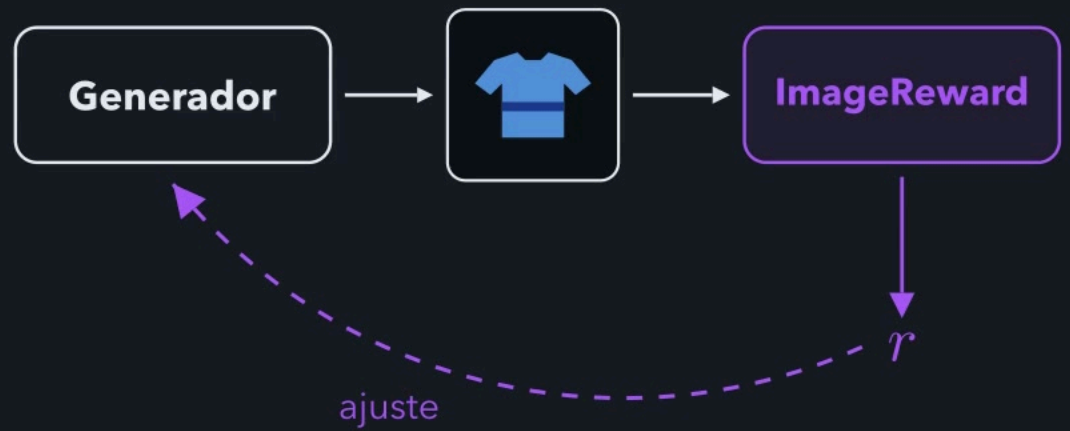
Cinco conceptos base

1 Difusión latente | 2 CLIP | 3 IP-Adapter | 4 Recomendadores | 5 Reward feedback

Recomendadores



Reward feedback learning



Zhou et al. (2018) · Kang & McAuley (2018) · He et al. (2017) · Xu et al. (2023)

2 Marco teórico

Trabajos relacionados

Difusión
instrucción explícita

- Textual Inversion
- DreamBooth

LLM
historial → texto

- PMG
- RAGAR

MLLM
features visuales

- LaVIT
- I-AM-G
- Pigeon

historial → features → generador

supuesto: **el historial basta**

Gal et al. (2023) · Ruiz et al. (2023) · Shen et al. (2024) · Ling et al. (2025) · Jin et al. (2024) · Wang et al. (2024) · Xu et al. (2025)

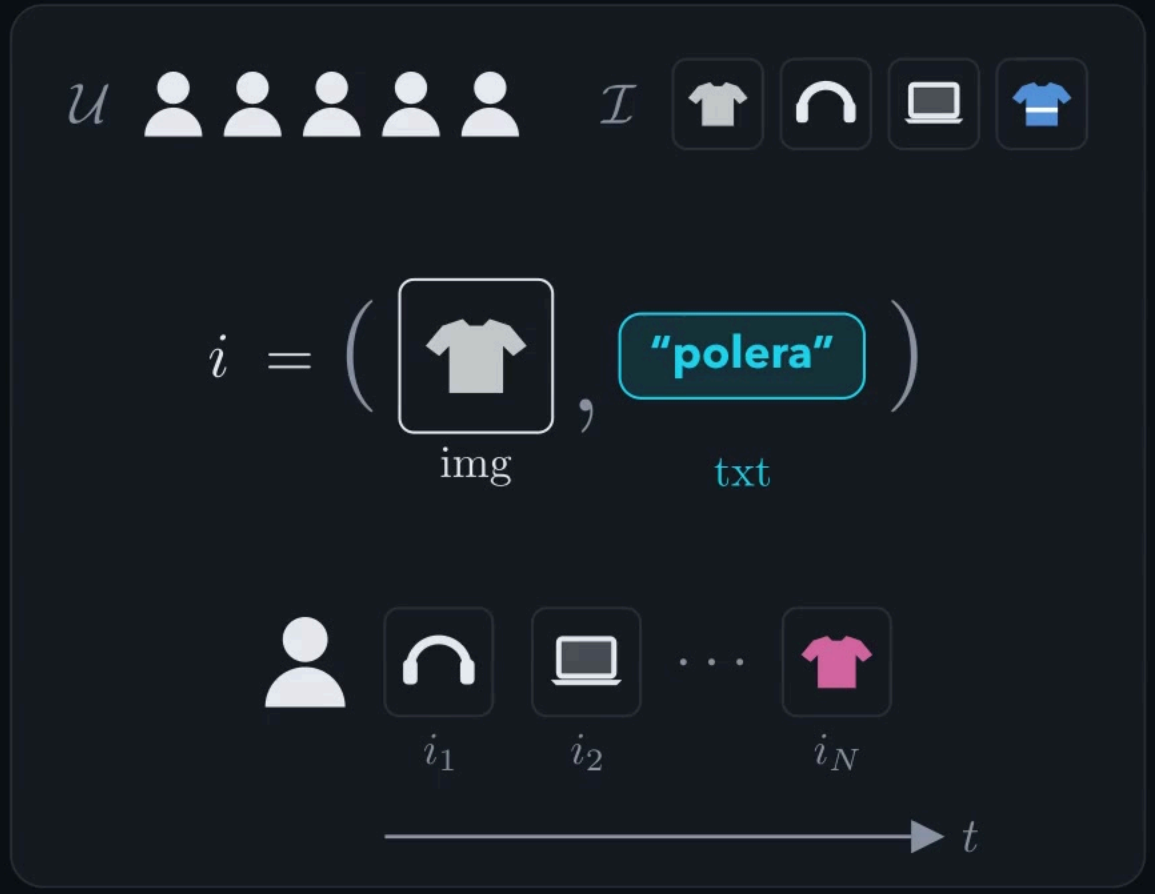
3 Problema

Formulación del problema

$\mathcal{U},\ \mathcal{I}$ usuarios, ítems

$i = (\text{img}, \text{txt}) \in \mathcal{I}$ imagen + texto

$S_u = \{i_1, i_2, \ldots, i_N\}$ orden temporal



Zhang et al. (2026), Sec. 3.1

3 Problema

Formulación del problema

$\mathcal{U},\ \mathcal{I}$ — usuarios, ítems

$i = (\text{img}, \text{txt}) \in \mathcal{I}$ — imagen + texto

$S_u = \{i_1, i_2, \ldots, i_N\}$ — orden temporal

$|S_u| = N \leq m$ **m = 3** — usuario sparse



Zhang et al. (2026), Sec. 3.1

3 Problema

Formulación del problema

$\mathcal{U},\ \mathcal{I}$ usuarios, ítems

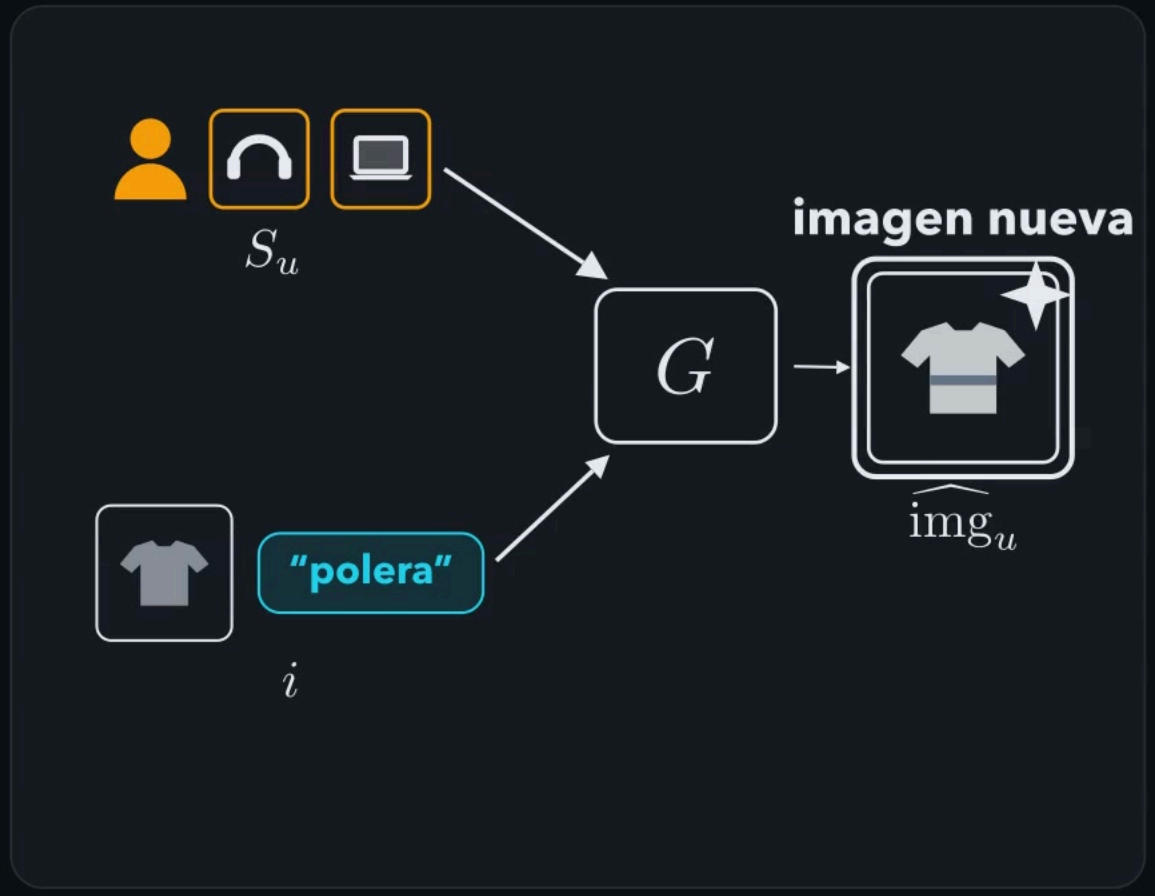
$i = (\text{img}, \text{txt}) \in \mathcal{I}$ imagen + texto

$S_u = \{i_1, i_2, \ldots, i_N\}$ orden temporal

$|S_u| = N \leq m$ **m = 3** usuario sparse

$$\widehat{\text{img}}_u = G(S_u, i)$$

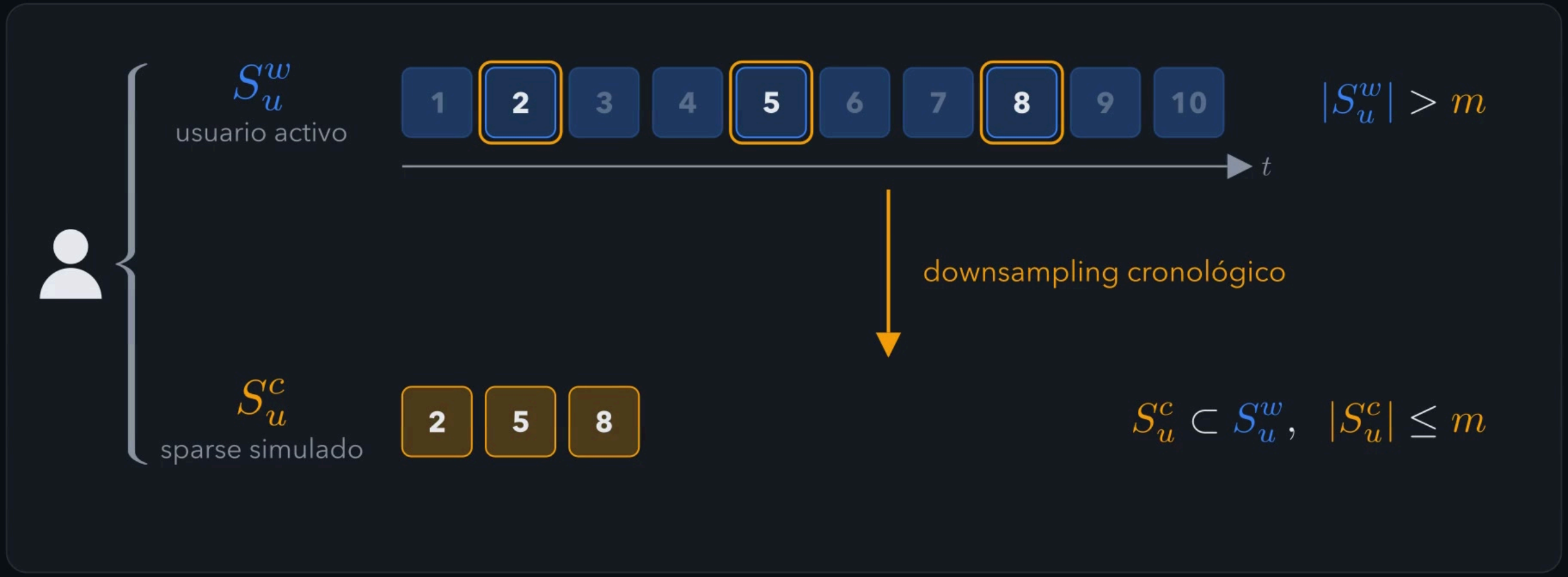
✓ **Personalización** ✓ **Alineación semántica**



Zhang et al. (2026), Sec. 3.1

3 Problema

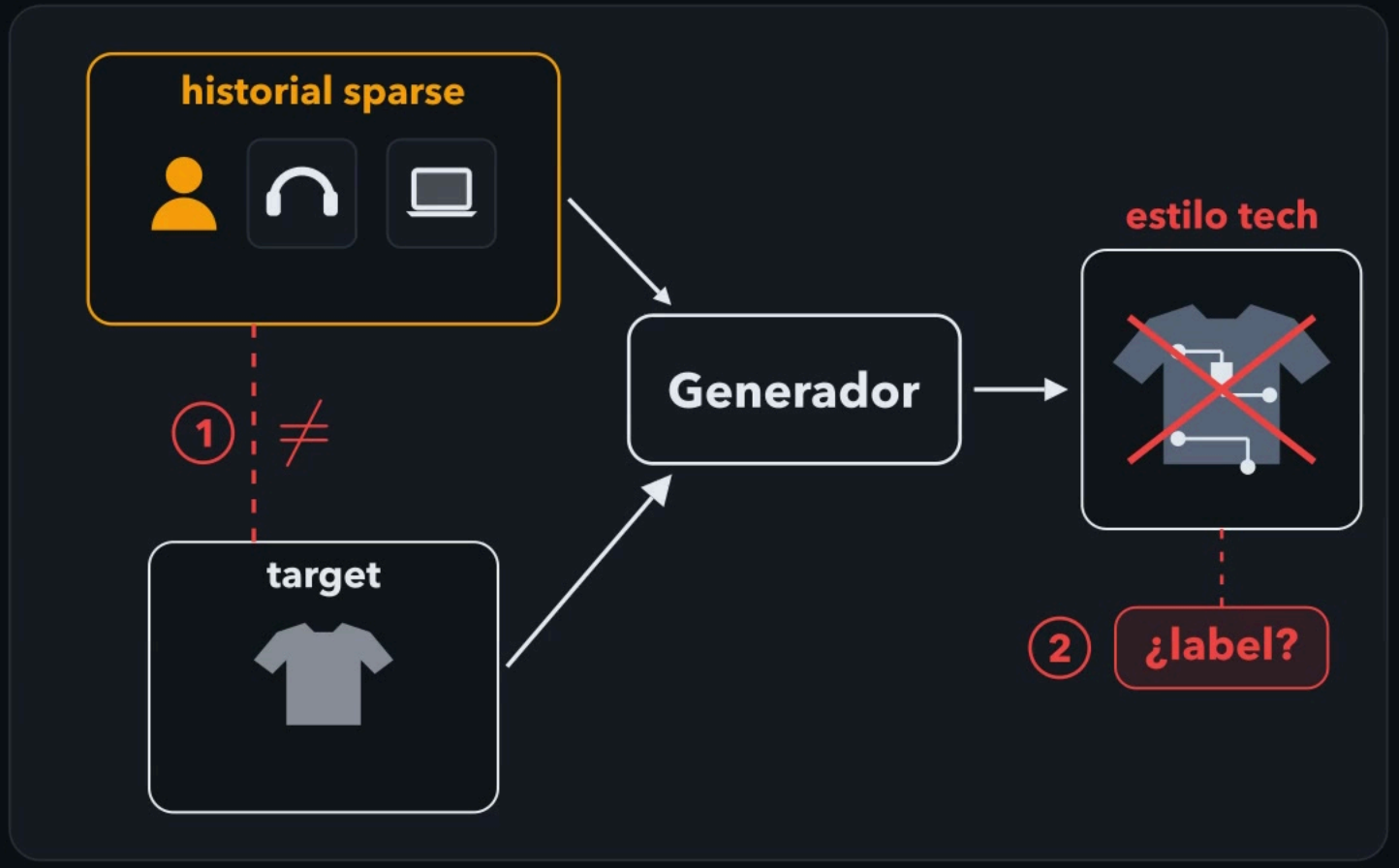
Entrenamiento: sparse simulado



Zhang et al. (2026), Sec. 3.1 y 4.1.1

3 Problema · Fig. 1

¿Por qué es difícil?



1 Desalineamiento de preferencias

2 Falta de supervisión

$$\min\ d(\widehat{\text{img}},\ \text{img})$$

previos: optimiza lo observado

Zhang et al. (2026), Sec. 1 y Fig. 1 (recreada)

4 Contribución

¿Qué aporta UDSIG?

	Preferencias implícitas	Referencias alineadas al target	Supervisión de preferencias confiable	Diseñado para sparse
TI / DreamBooth	—	—	—	—
PMG	✓	—	—	—
RAGAR	✓	—	◐	—
Pigeon	✓	—	—	—
UDSIG	✓	✓	✓	✓

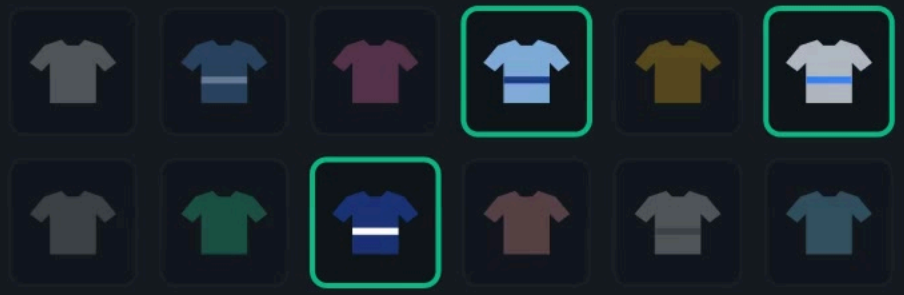
✓ sí ◐ parcial — no

Zhang et al. (2026), Sec. 1-2 · Gal et al. (2023) · Ruiz et al. (2023) · Shen et al. (2024) · Ling et al. (2025) · Xu et al. (2025)

4 Contribución

Dos ideas

Buscar en vez de inyectar



todo el catálogo

Aprender de quien sabe

Profesor → reward → **Generador**

S_u^w dense

S_u^c sparse

Zhang et al. (2026), Sec. 1

4 Contribución

Contribuciones

1 **Framework sparse**
primero, según los autores

2 **Retrieval de baja incertidumbre**

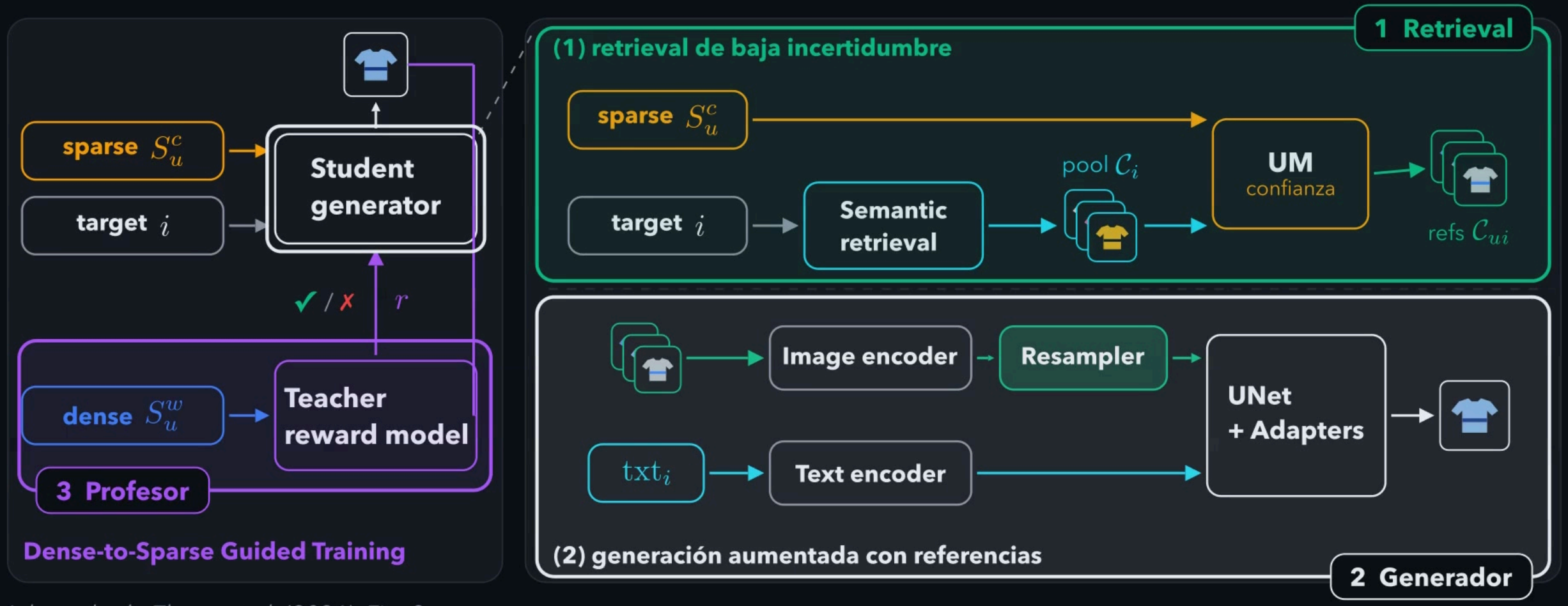
3 **Entrenamiento dense-to-sparse**

4 **3 datasets + evaluación humana**

Zhang et al. (2026), Sec. 1

5 Visión general

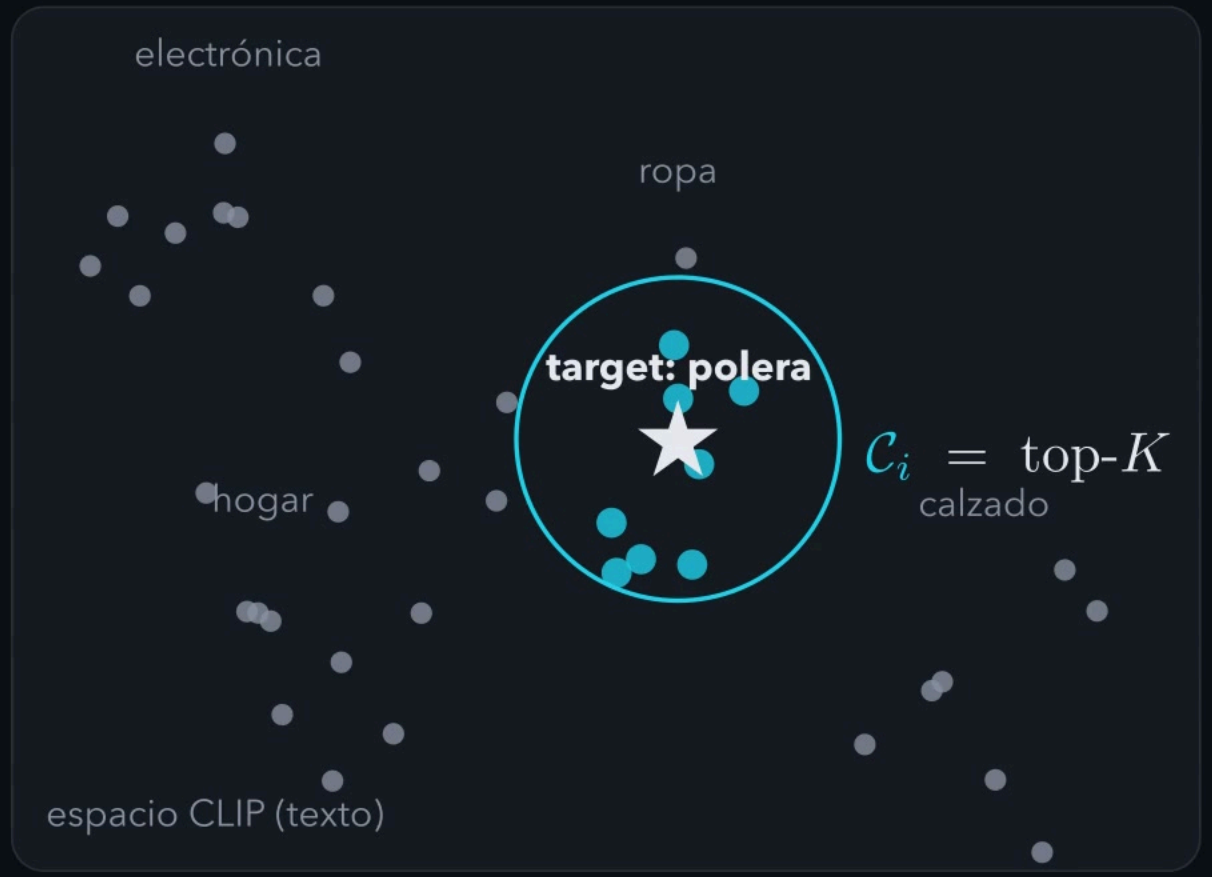
UDSIG: el framework completo



Adaptado de Zhang et al. (2026), Fig. 2

5.1 SLR

Semantic-aware Low-uncertainty Retrieval



$$E_j = \text{Encoder}(\text{txt}_j) \quad (1)$$

$$s_{ij} = \frac{E_i^\top E_j}{\|E_i\| \, \|E_j\|} \quad (2)$$

$$\mathcal{C}_i = \text{top-}K$$

Zhang et al. (2026), Eq. 1-4 · Radford et al. (2021) · Zhou et al. (2018)

5.1 SLR

Semantic-aware Low-uncertainty Retrieval



$$E_j = \text{Encoder}(\text{txt}_j) \quad (1)$$

$$s_{ij} = \frac{E_i^\top E_j}{\|E_i\| \, \|E_j\|} \quad (2)$$

$$d_{uj} = \text{UM}(u, S_u^c, j) \in (0, 1) \quad (3)$$

Zhang et al. (2026), Eq. 1-4 · Radford et al. (2021) · Zhou et al. (2018)

5.1 SLR

Semantic-aware Low-uncertainty Retrieval



$$E_j = \mathrm{Encoder}(\mathrm{txt}_j) \quad (1)$$

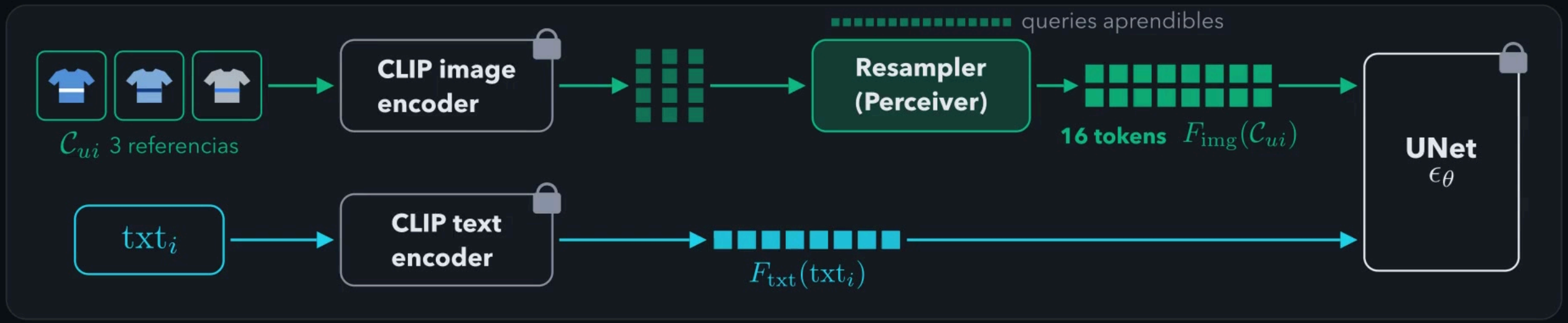
$$s_{ij} = \frac{E_i^\top E_j}{\|E_i\| \, \|E_j\|} \quad (2)$$

$$d_{uj} = \mathrm{UM}(u, S_u^c, j) \in (0, 1) \quad (3)$$

$$\mathcal{C}_{ui} = \mathrm{ArgTop}_k \{ d_{uj} \cdot s_{ij} \mid j \in \mathcal{C}_i \} \quad (4)$$

Zhang et al. (2026), Eq. 1-4 · Radford et al. (2021) · Zhou et al. (2018)

5.2
Reference Augmented Sparse Generation

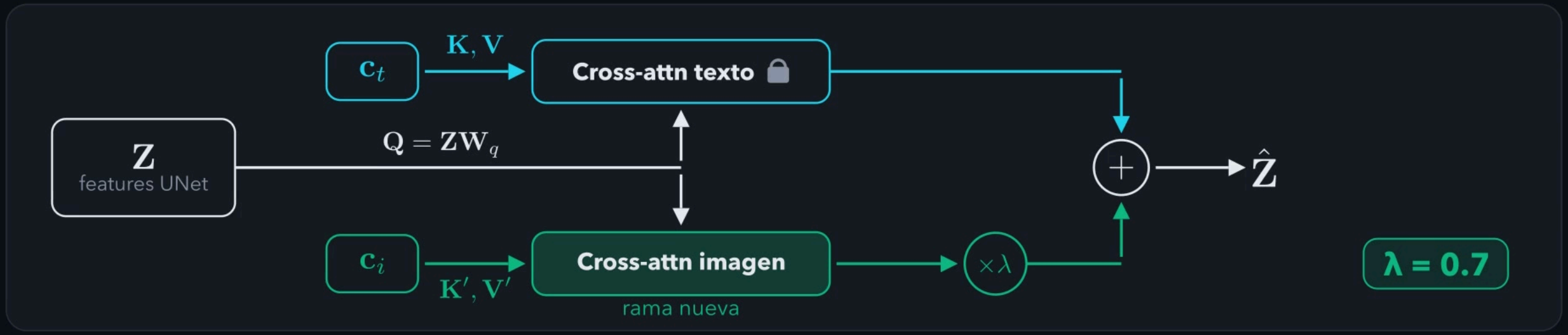


$$\epsilon_\theta\left(\mathbf{x}_t,\ F_{\text{img}}(\mathcal{C}_{ui}),\ F_{\text{txt}}(\text{txt}_i),\ t\right) \quad (5)$$

Zhang et al. (2026), Eq. 5-6, Tab. 1 · Ye et al. (2023) · Jaegle et al. (2021) · Radford et al. (2021)

5.2
Reference Augmented Sparse Generation

$$\epsilon_\theta\left(\mathbf{x}_t,\ F_{\text{img}}(C_{ui}),\ F_{\text{txt}}(\text{txt}_i),\ t\right) \quad (5)$$



$$\hat{\mathbf{Z}} = \underbrace{\text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}} + \lambda \cdot \underbrace{\text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}'^\top}{\sqrt{d}}\right)\mathbf{V}'} \quad (6)$$

Zhang et al. (2026), Eq. 5-6, Tab. 1 · Ye et al. (2023) · Jaegle et al. (2021) · Radford et al. (2021)

5.2

Reference Augmented Sparse Generation

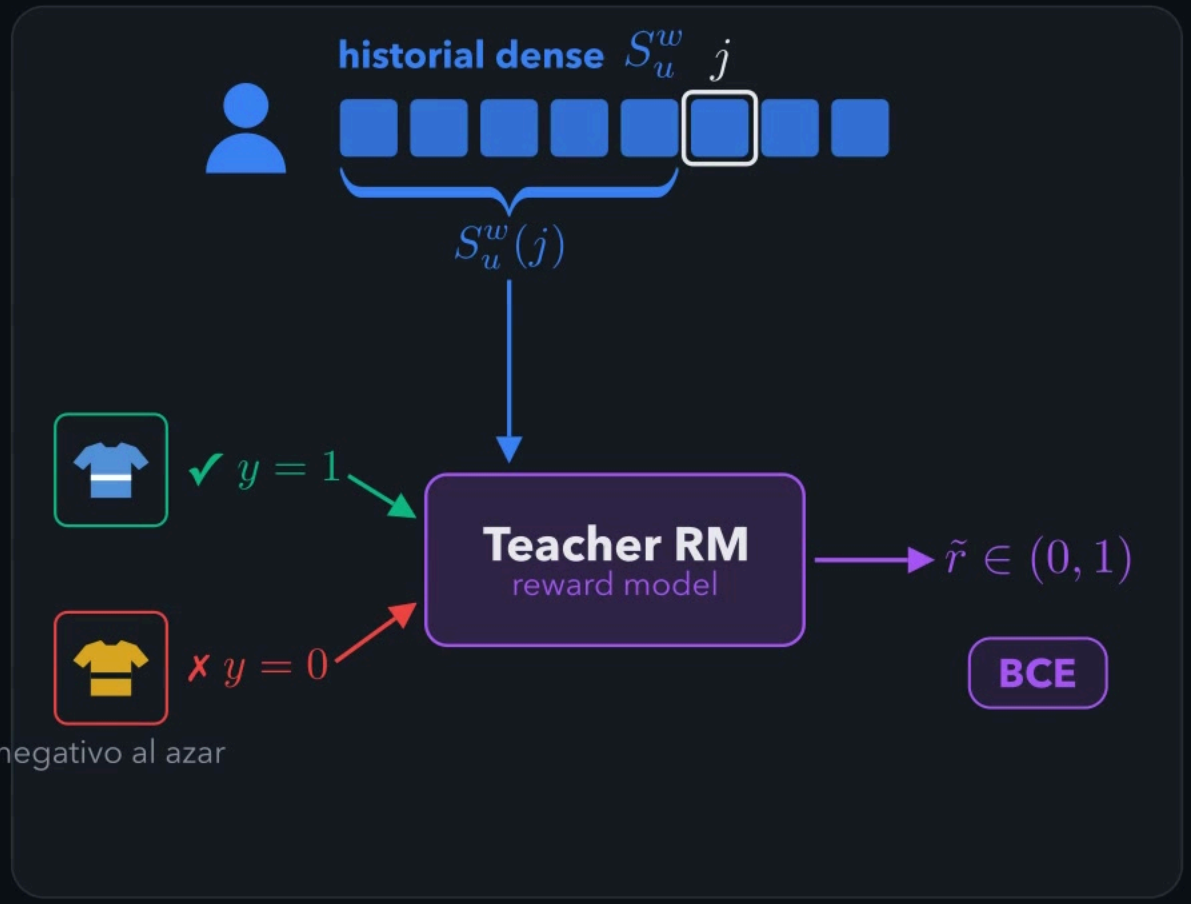


~3 h de entrenamiento **1× GPU A100**

Zhang et al. (2026), Eq. 5-6, Tab. 1 · Ye et al. (2023) · Jaegle et al. (2021) · Radford et al. (2021)

5.3 D2S

Dense-to-Sparse Guided Training



$$\tilde{r}_{u,\widetilde{img}_j} = \mathrm{RM}(u,\, S_u^w(j),\, \widetilde{img}_j) \tag{7}$$

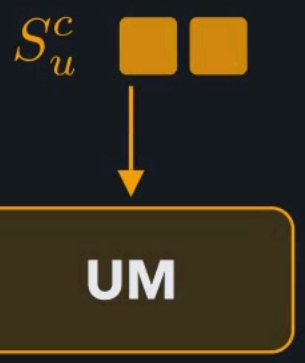
$$\mathcal{L}_{\mathrm{RM}} = \mathbb{E}_{u\sim\mathcal{U}}\, \mathbb{E}_{j\sim S_u^w}\, \mathrm{BCE}(\tilde{r}_{u,\widetilde{img}_j},\, y_{u,\widetilde{img}_j})$$

Zhang et al. (2026), Eq. 7-11 · Xu et al. (2023) ImageReward · Zhou et al. (2018)

5.3 D2S

Dense-to-Sparse Guided Training

UM estimador
retrieval

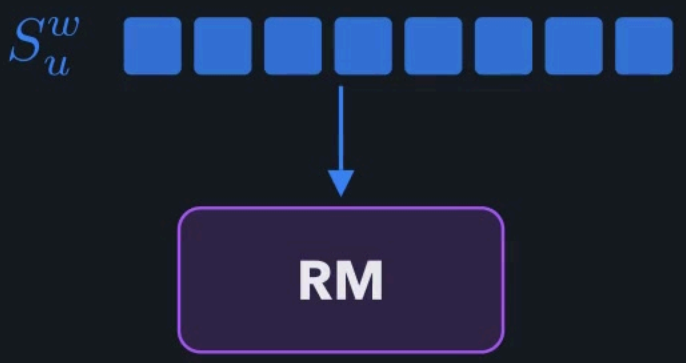


elige referencias

$\neq$

instancias independientes

RM profesor
supervisión

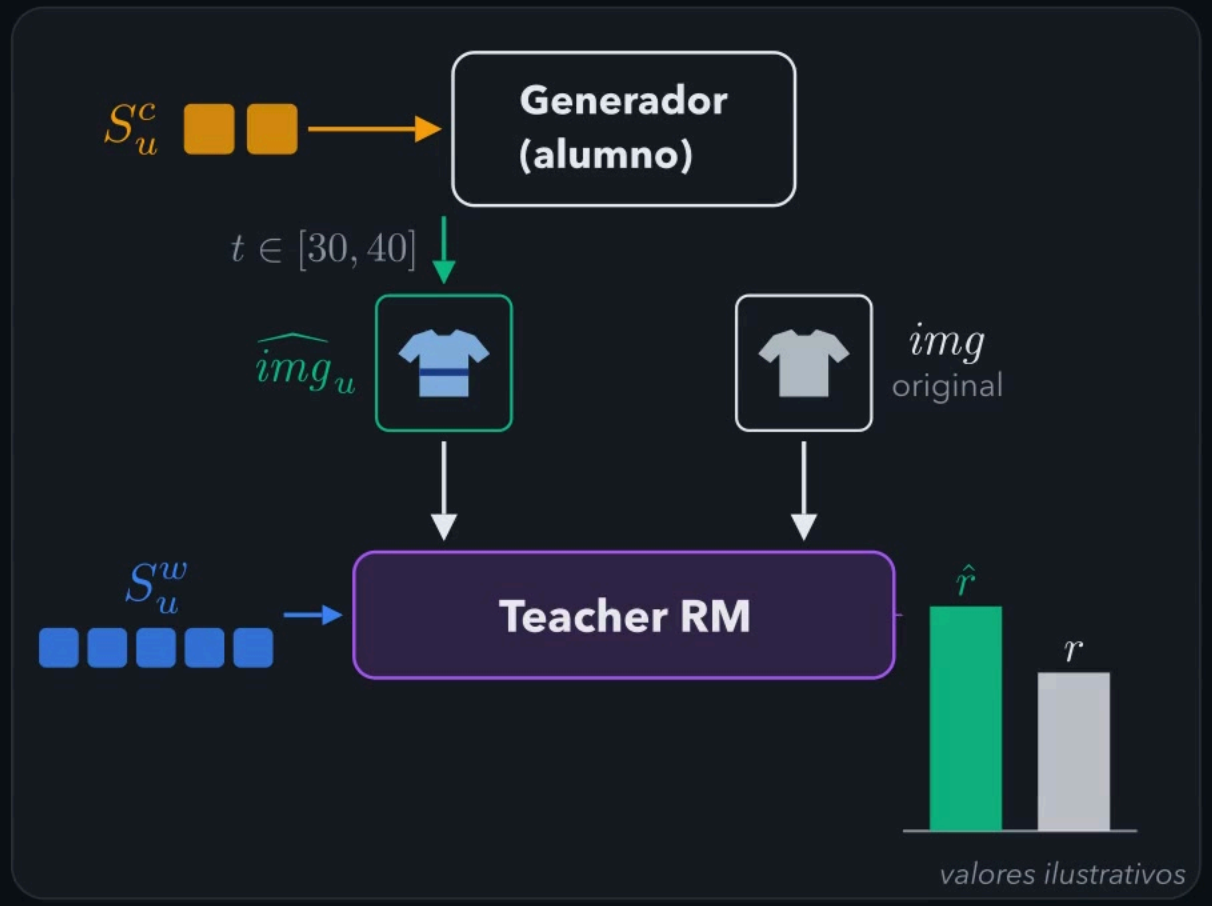


puntúa imágenes

Zhang et al. (2026), Eq. 7-11 · Xu et al. (2023) ImageReward · Zhou et al. (2018)

5.3 D2S

Dense-to-Sparse Guided Training



$$r_{u,\widehat{img}_u} = \mathrm{RM}(u,\, S_u^w,\, \widehat{img}_u) \qquad (8)$$

$$r_{u,img} = \mathrm{RM}(u,\, S_u^w,\, img)$$

$$\mathcal{L}_r = \mathbb{E}_{(u,i)\sim\mathcal{D}}\Big(\phi\big(r_{u,\widehat{img}_u} - r_{u,img}\big)\Big) \qquad (9)$$

$$\phi(x) = 1 - x$$

$$\hat{r} > r$$

Zhang et al. (2026), Eq. 7-11 · Xu et al. (2023) ImageReward · Zhou et al. (2018)

5.3 D2S

Dense-to-Sparse Guided Training

$$\mathcal{L}_r = \mathbb{E}_{(u,i)\sim\mathcal{D}}\Big(\phi\big(r_{u,\widehat{img}_u} - r_{u,img}\big)\Big) \quad (9)$$

reward del profesor

$$\mathcal{L}_p = \mathbb{E}_{(u,i)\sim\mathcal{D}}\Big(\mathbb{E}_{\epsilon\sim\mathcal{N}(0,1),\,t}\big\|\epsilon - \epsilon_\theta\big(\mathbf{x}_t,\, F_{\text{img}}(\mathcal{C}_{ui}),\, F_{\text{txt}}(\text{txt}_i),\, t\big)\big\|_2^2\Big) \quad (10)$$

denoising estándar

$$\mathcal{L} = \alpha\,\mathcal{L}_r + \mathcal{L}_p \quad (11)$$

α = 0.8

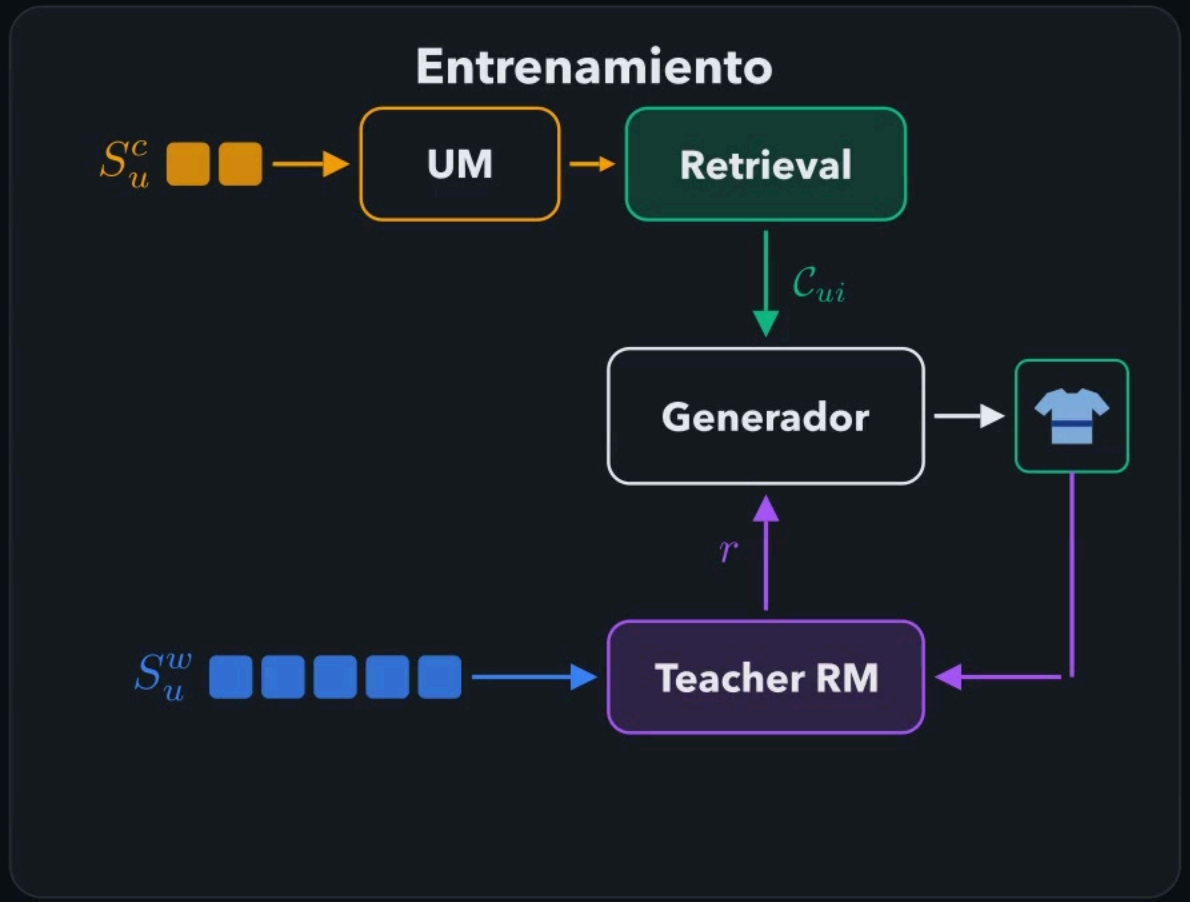
$$\text{train} \cap \text{test} = \varnothing$$

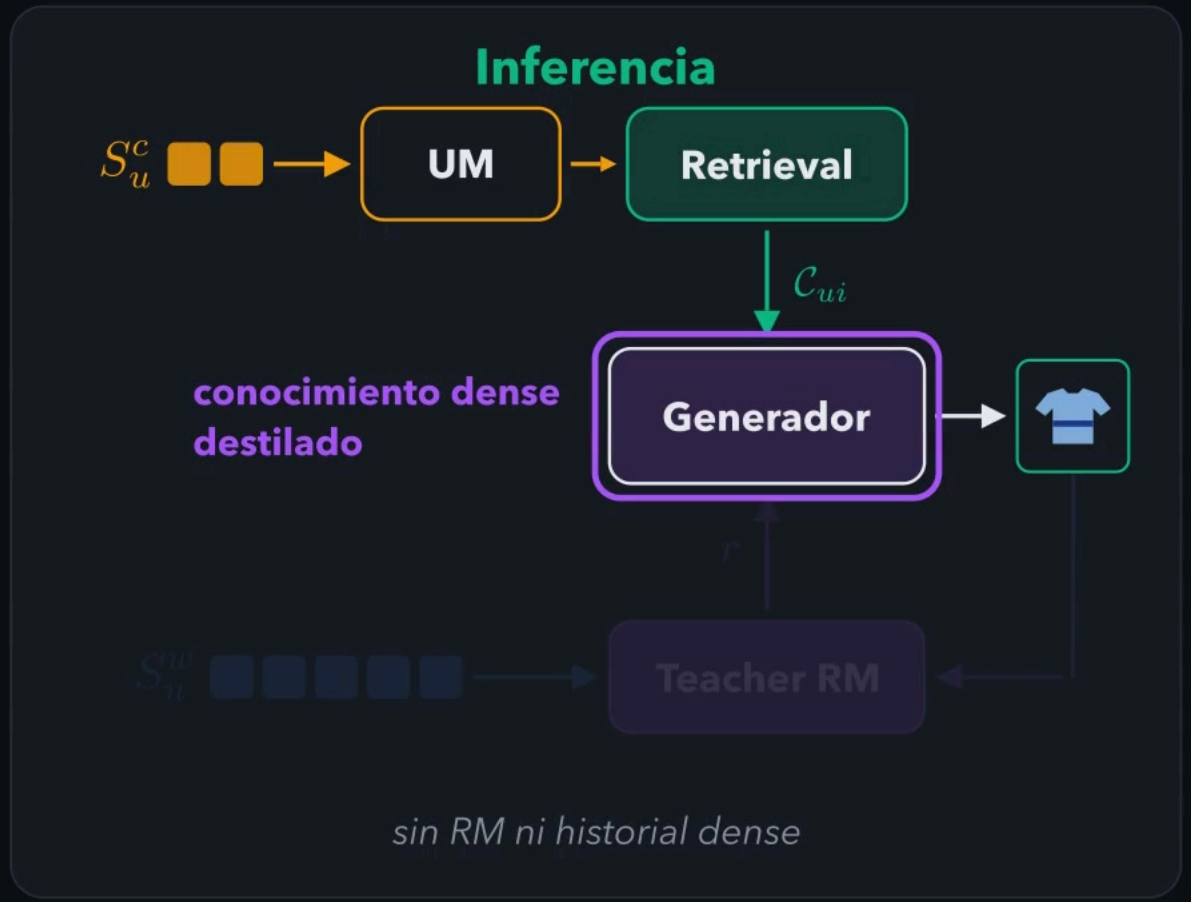


Zhang et al. (2026), Eq. 7-11 · Xu et al. (2023) ImageReward · Zhou et al. (2018)

5.4

Entrenamiento vs inferencia





Zhang et al. (2026), Sec. 3.4

6.1

Setup experimental

POG
moda

1.000 usuarios
14.582 ítems

SER
stickers

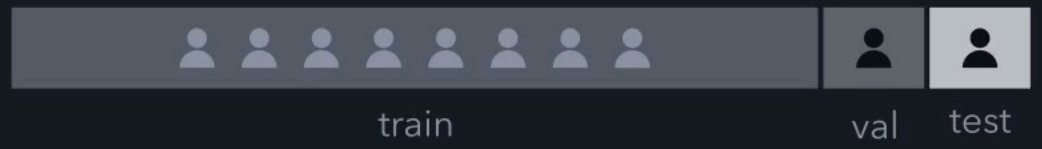
14.345 ítems

MovieLens
películas

598 usuarios
6.041 ítems

rating ≥ 4

split por usuario $8:1:1$



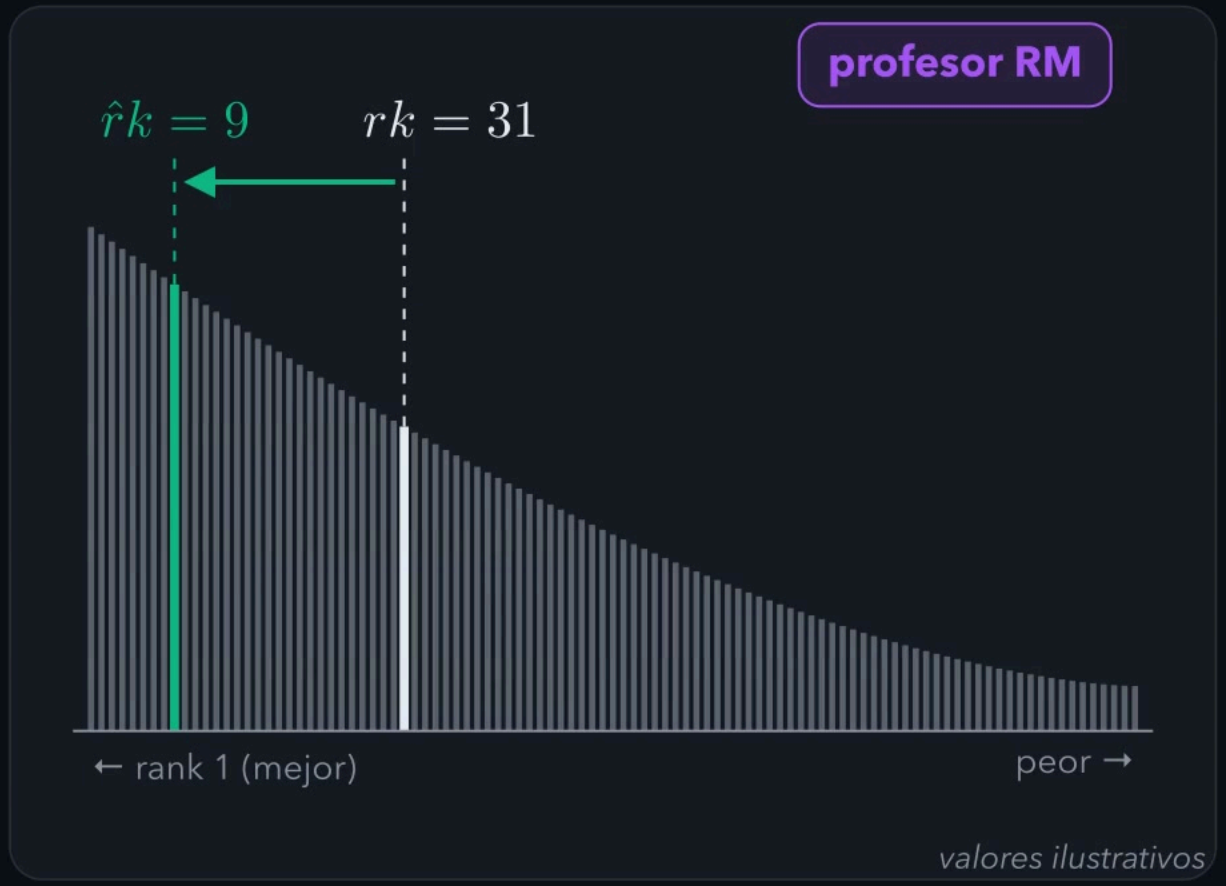
sparse simulado: 1-3 interacciones

S_u^c $|S_u^c| \leq m = 3$ t

Zhang et al. (2026), §4.1.1 · Gao et al. (2022), simulación sparse

6.1

Métrica central: ΔR



$$\Delta R = \frac{rk - \hat{r}k}{1 + \mathrm{Num}} \times 100\%$$

$$\mathrm{Num} = 99$$

$$\frac{31 - 9}{1 + 99} \times 100\% = +22\%$$

ΔR ↑ mayor es mejor

ΔR > 0: la generada supera a la original

Zhang et al. (2026), §4.1.3 · Wu et al. (2022), protocolo de negativas

6.1

Métricas y baselines

Personalización	ΔR ↑
Semántica CLIP · DINO	CS ↑ · CIS ↑ · DIS ↑
Perceptual	LPIPS ↓ · MS-SSIM ↑
Fidelidad	FID ↓

8 baselines

Diffusion	LLM	MLLM
TI	PMG	Qwen3-VL
DreamBooth	RAGAR	LaVIT
		I-AM-G
		Pigeon

Zhang et al. (2026), §4.1.3-4.1.4 · Radford et al. (2021) · Oquab et al. (2024) · Heusel et al. (2017)

6.2 Tab. 2

Resultados principales



Zhang et al. (2026), Tab. 2

6.2 Tab. 2

Resultados principales



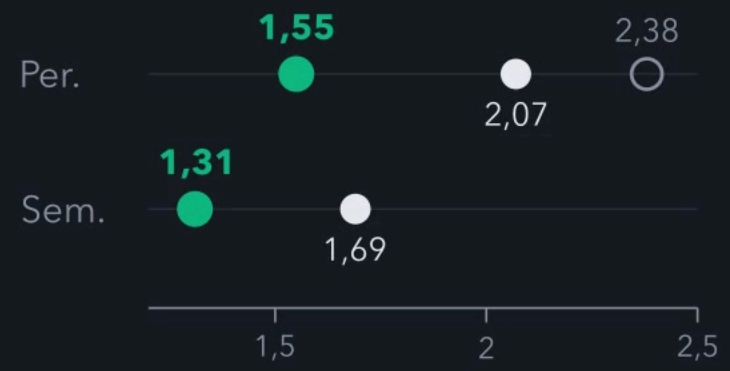
Zhang et al. (2026), Tab. 2

6.3 Tab. 3

Evaluación humana

rank promedio ↓ (1 = mejor) ● **UDSIG** ● Pigeon ○ ORI

POG



SER



MovieLens



hasta **+25,12%** personalización
POG, vs Pigeon

hasta **+24,56%** semántica
SER, vs Pigeon

50 participantes
50 casos por dataset

Zhang et al. (2026), Tab. 3 y §4.3

6.4

Ablation sobre POG (Tab. 4)

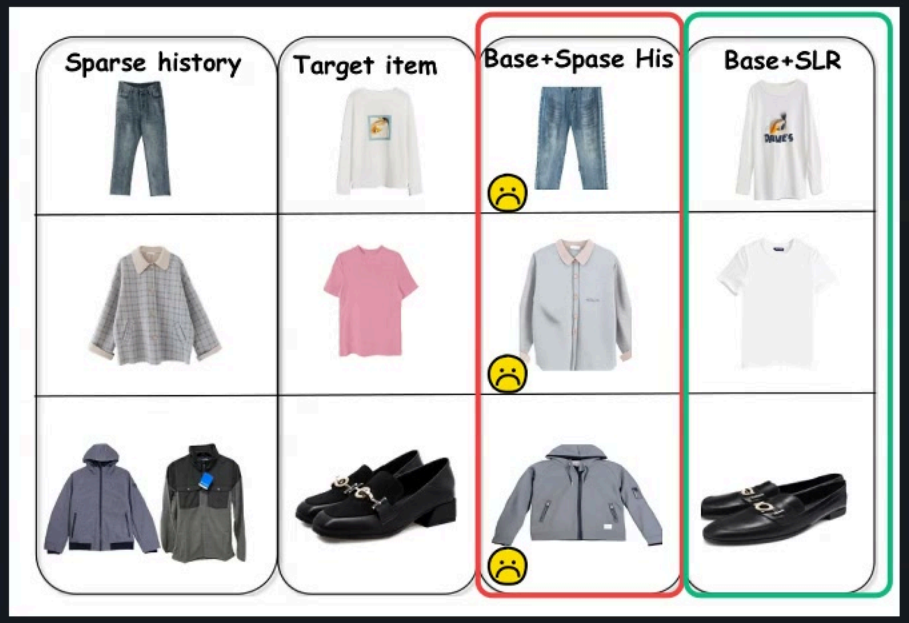
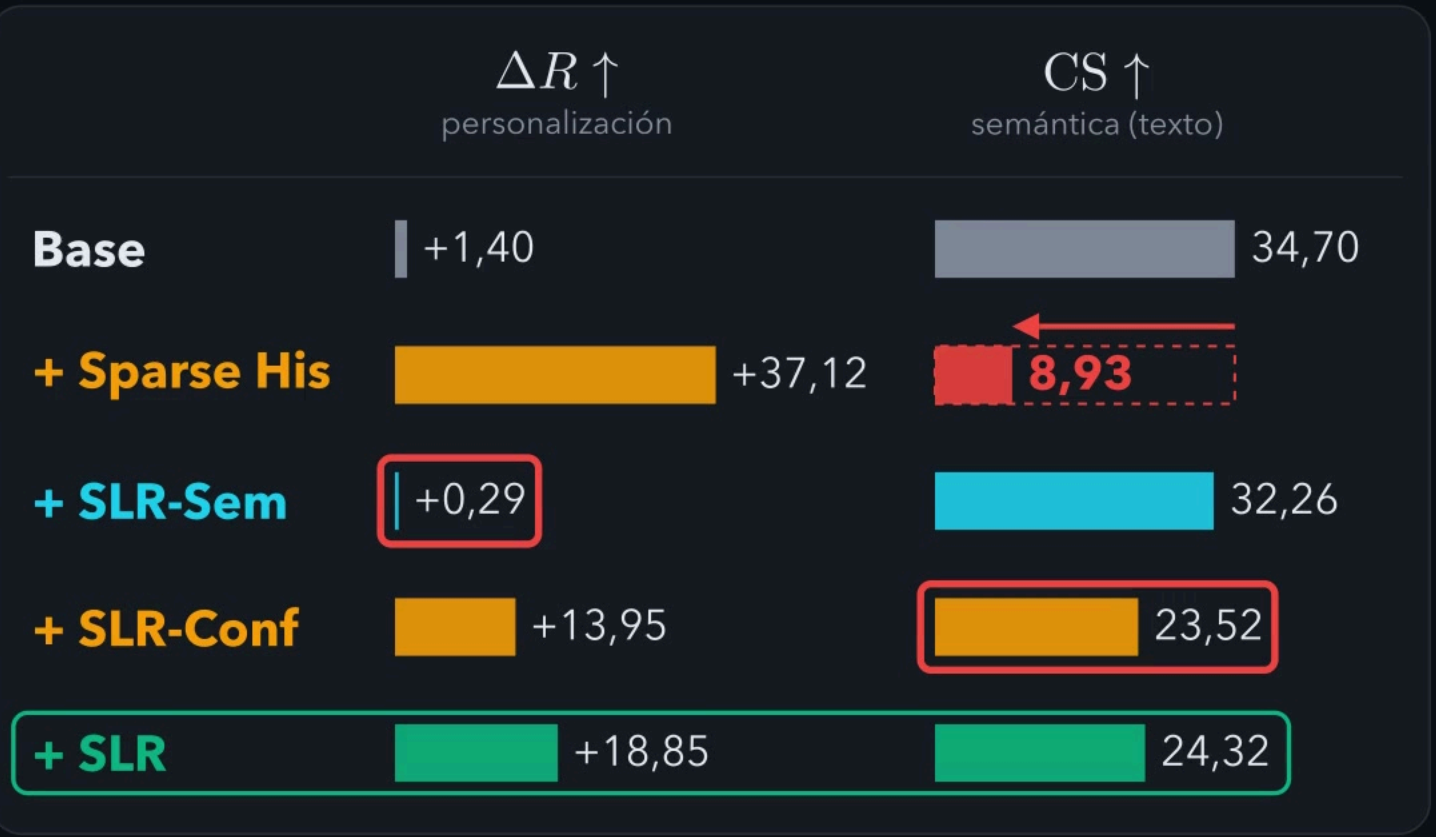


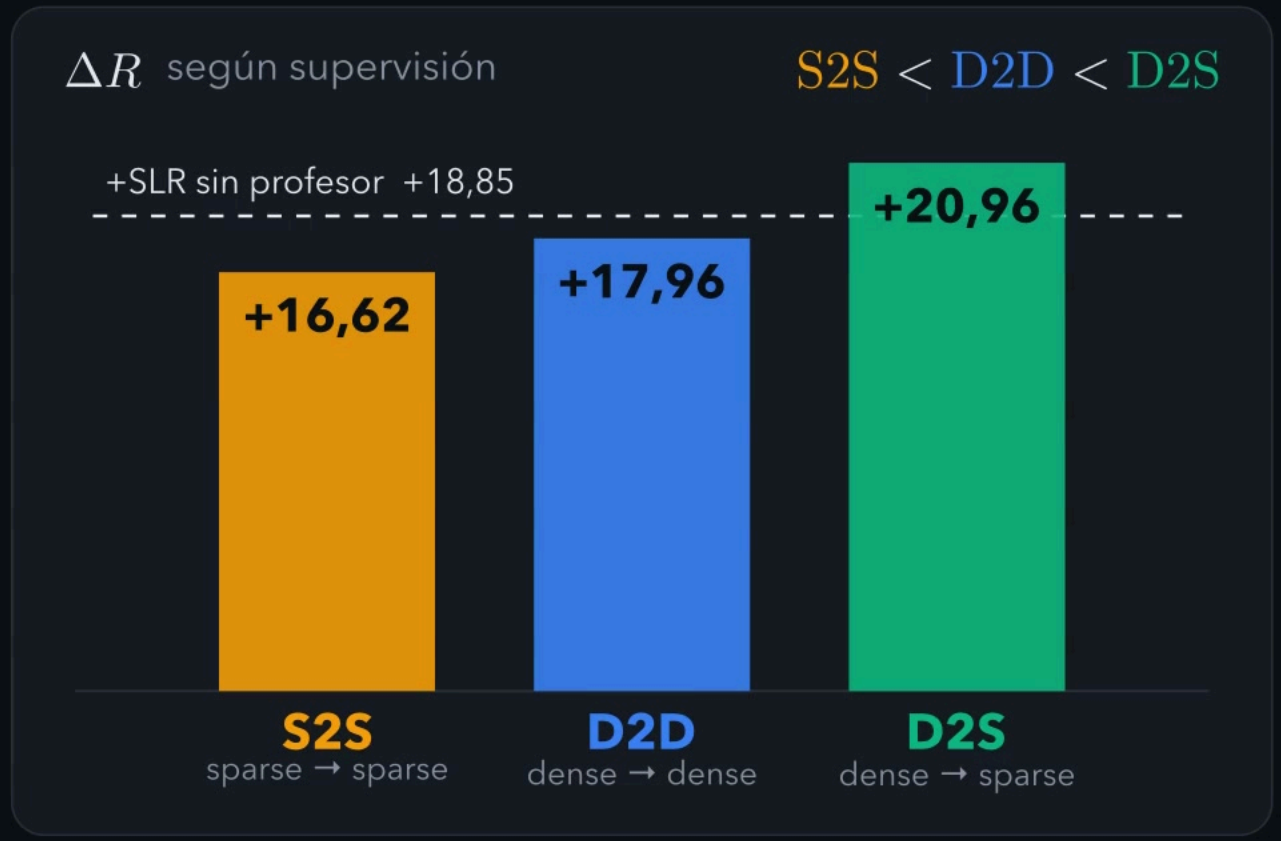
Fig. 3



Fuente: Zhang et al. (2026), Tab. 4 y Fig. 3, CC BY 4.0

6.4

Ablation sobre POG (Tab. 4)



Backbone · ΔR

	UM +SLR	RM +SLR+D2S
DIN	+18,85	+20,96
SASRec	+18,74	+20,85
NeuMF	+17,06	+19,84

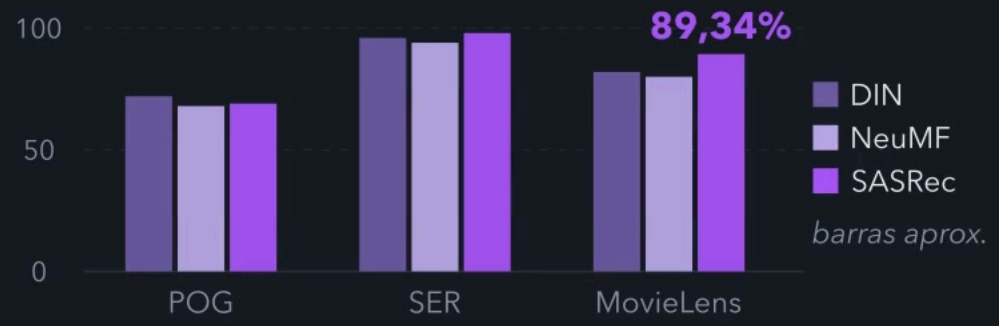
$$\text{DIN} \approx \text{SASRec} \approx \text{NeuMF}$$

Fuente: Zhang et al. (2026), Tab. 4 y Fig. 3, CC BY 4.0

6.5

Análisis adicional

Profesor confiable pair accuracy · Fig. 4

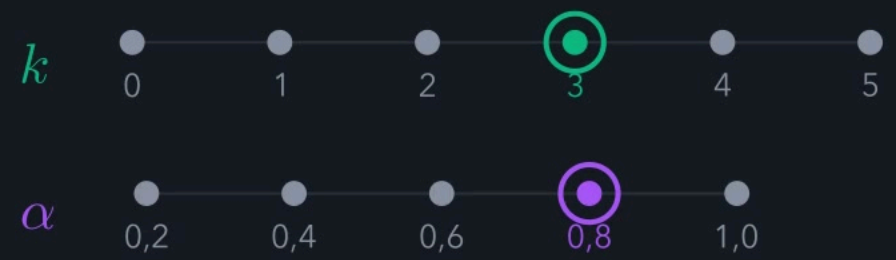


Generalización Fig. 5

dense $|S_u| \geq 10$ UDSIG > PMG, Pigeon

plug-in SLR PMG ↑ LaVIT ↑

Hiperparámetros Fig. 6



Eficiencia 1× A100 · Sec. 4.1.2

UDSIG ≈ 4 s / imagen

Pigeon 16 s

entrenamiento **≈ 3 h**

Zhang et al. (2026), Fig. 4-6 y Sec. 4.1.2, 4.4-4.6

6.6

Caso cualitativo (POG)



score RM

caso 2

original
0,26

Pigeon
0,28

UDSIG
0,41

Fuente: Zhang et al. (2026), CC BY 4.0

6.6

Limitaciones

Sparse simulado	downsampling offline · reconocido en el paper
Juez ≈ entrenador	RM supervisa y calcula ΔR · **evaluación humana compensa en parte**
RAGAR en SD 1.5	resto en SDXL
Sin cold start puro	$\lvert S_u \rvert = 0$ fuera de alcance

Zhang et al. (2026), Sec. 4.1.2 y 5 · lectura crítica propia

Cierre

Conclusión

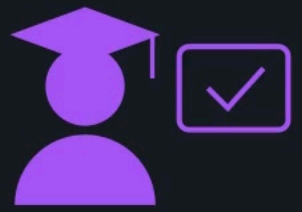
1



Buscar > inyectar

con historial sparse

2



Profesor dense

supervisa sin labels

3



Piezas reutilizables

SLR como plug-in

Formato ACM

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</> Código: `github.com/zhangyuting725/UDSIG`